H1

ANDERSON SERANGOON JUNIOR COLLEGE
HIGHER 1

2021 JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME	

CLASS	2	1	/			INDEX NUMBER	

BIOLOGY 8876/02

Paper 2 Structured and Free Response Questions

15 September 2021
Wednesday

Candidates answer on the Question Paper.
No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Total (P1)	**/ 30**
Section A	
1	
2	
3	
4	
5	
Section B	
6 or 7	
Total (P2)	**/ 60**
100%	
Grade	

This document consists of **20** printed pages.

Section A starts on page 3

Section A

Answer **all** the questions.

1 Microorganisms and their associated enzymes are economically important, and are widely used in various industries.

(a) For a commercial application using an enzyme, the progress of the enzyme-catalysed reaction needs to be studied.

Outline how the progress of an enzyme-catalysed reaction can be investigated experimentally.

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The enzyme lactase is commercially extracted from the unicellular fungus *Kluyveromyces lactis*, which is naturally found in dairy products. Lactase catalyses the breakdown of lactose, a sugar found in milk, and is used to produce lactose-free milk.

The reaction catalysed by lactase is summarised in Fig. 1.1.

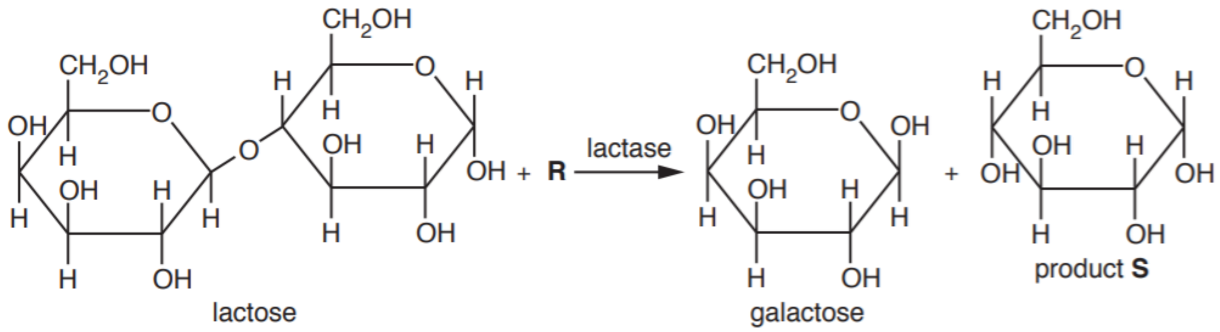

Fig. 1.1

(b) Describe the reaction that is catalysed by lactase. Use Fig. 1.1 to help you.

In your answer, identify **R** and product **S**.

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In the syrup industry, the starch extracted from barley seeds (barley starch) is used in the production of sugar syrups. One key enzyme synthesised by the barley seed is α-amylase, which breaks down stored starch during seedling formation.

Fig. 1.2 is a graph showing how the activity of α-amylase extracted from barley seeds changes as the temperature increases from 10 °C to 66 °C.

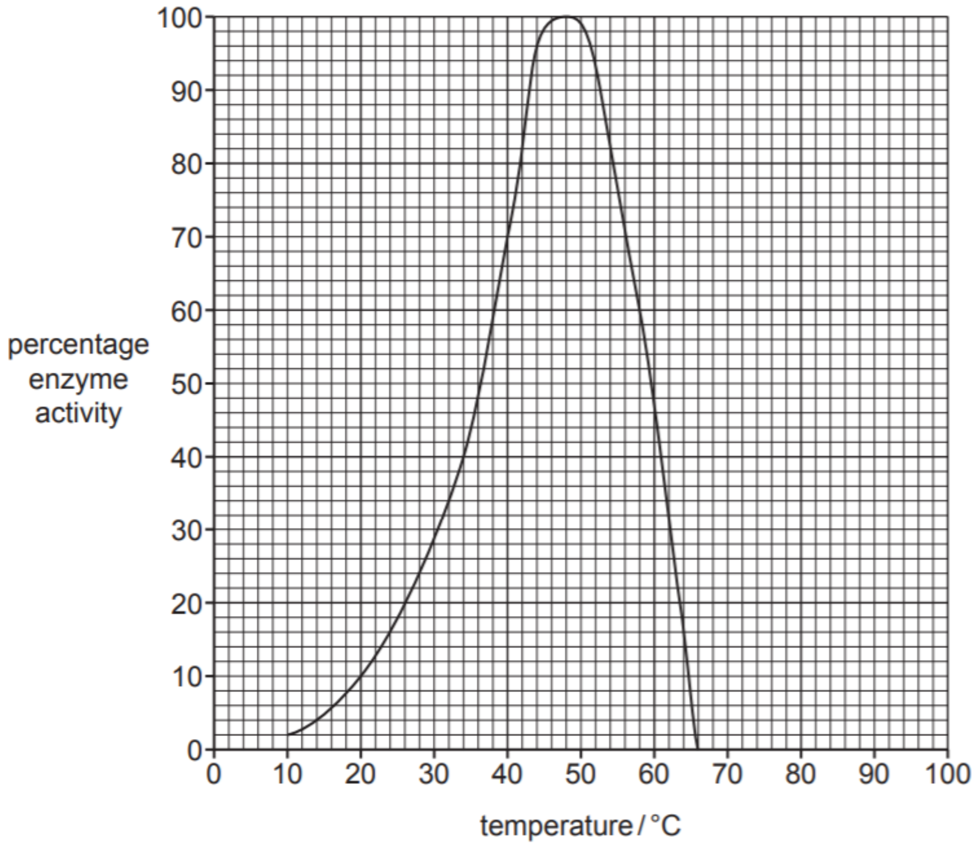

Fig. 1.2

(c) Explain the effect of temperature on the activity of α-amylase extracted from barley seeds, as shown in Fig. 1.2.

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(d) When producing sugar syrups, there are advantages in using enzymes extracted from microorganisms, rather than enzymes extracted from barley seeds.

For example, some enzymes extracted from microorganisms are heat stable. Heat-stable enzymes are used to increase productivity because the reactions can be carried out at higher temperatures.

Sketch on Fig. 1.2 the curve that would be obtained using α-amylase enzyme that is heat stable.

[1]

Microorganisms, specifically bacteria, are also used in the mining industry.

When gold is associated with mineral ores such as iron sulfide, the sulfides must be oxidised to release the gold particles.

Since the mid 1990s, gold has been extracted from such ores by bioleaching. Bioleaching involves the use of suitable bacteria such as *Acidithiobacillus ferrooxidans*, which oxidise iron sulfide to soluble iron sulfate, releasing Fe^{3+} and SO_4^{2-} ions.

Gold-bearing sulfide ores often contain arsenic, which is potentially toxic to the bacteria used in bioleaching. However, arsenic-resistant strains of *A. ferrooxidans* have been found in some mines.

The activity of two strains of the bacterium, in the presence and absence of arsenic ions, is shown in Table 1.1.

Table 1.1

	oxidation rate of iron in the ore / mg dm^{-3} h^{-1}	
strain of *A. ferrooxidans*	arsenic ions absent	arsenic ions present
1	16	15
2	48	47

(e) Describe the results shown in Table 1.1 **and** explain the role of natural selection in the evolution of arsenic-resistant bacteria.

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[Total: 13]

2 Fig. 2.1 shows the life cycle of the water flea (*Daphnia sp.*), which lives in various aquatic environments ranging from acidic swamps to freshwater lakes.

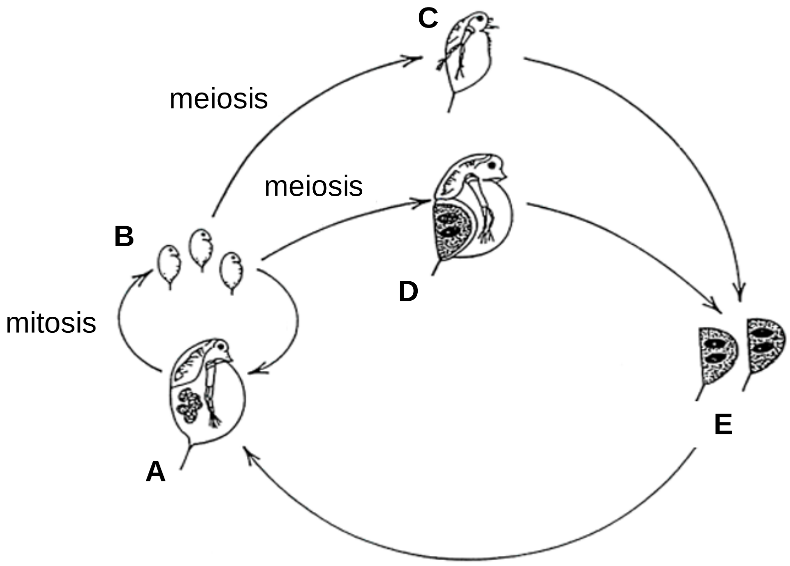

Fig. 2.1

- In favourable conditions, all the animals in a population are females (**A**).
- These females produce eggs by mitosis, which develop into young females (**B**) without being fertilised.
- In unfavourable conditions, eggs are produced by meiosis and develop directly without fertilisation into either males (**C**) or females (**D**).
- The eggs produced by females (**D**) are fertilised by sperm from the males (**C**), and then released in a protective egg case (**E**) which enables them to survive unfavourable conditions. When favourable conditions return, these fertilised eggs develop back into females (**A**).

(a) Explain why the eggs from **D** and the sperm from **C** must be produced by mitosis.

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(b) Explain why females (**A**) that developed from fertilised eggs (**E**) are genetically different from one other.

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(c) Biologists studied a population of water fleas in Felbrigg Hall, a shallow lake in England. The immediate region where the lake is located had experienced a threefold increase in the number of heat waves, and an average temperature rise of 1.15^{o}C since the 1960s.

The biologists reported that the water flea population has **not** declined since the 1960s, and has since adapted to the warmer lake conditions.

Using biological knowledge and the information provided including Fig. 2.1, comment on how the water flea population in Felbrigg Hall is likely to be different today, compared to the water flea population in the 1960s.

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[Total: 7]

3 The mitochondrion is an important organelle found in eukaryotic cells.

(a) A student made the following statement:

'Without the mitochondrion, cellular respiration cannot occur.'

Comment on the accuracy of the statement above.

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The mitochondrion contains its own DNA. Mitochondrial DNA (mtDNA) is a double-stranded molecule that consists of approximately 33,000 nucleotides.

(b) The two strands of mtDNA differ in their base composition, thus allowing both strands to be separated to give a heavy and a light strand by density centrifugation.

Suggest how the heavy strand differs from the light strand.

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The replication of mtDNA involves the Twinkle protein, which has both helicase and primase activity.

In humans, the Twinkle protein is encoded by the *TWNK* gene that is located in the long arm of chromosome 10.

(c) Fig. 3.1 is a simplified diagram of a human cell, which shows some key structures typically present.

On Fig. 3.1, use a labelling line and

(i) the letter **P** to label the precise location where transcription of the *TWNK* gene occurs. [1]

(ii) the letter **Q** to label the precise location where the Twinkle protein carries out its function. [1]

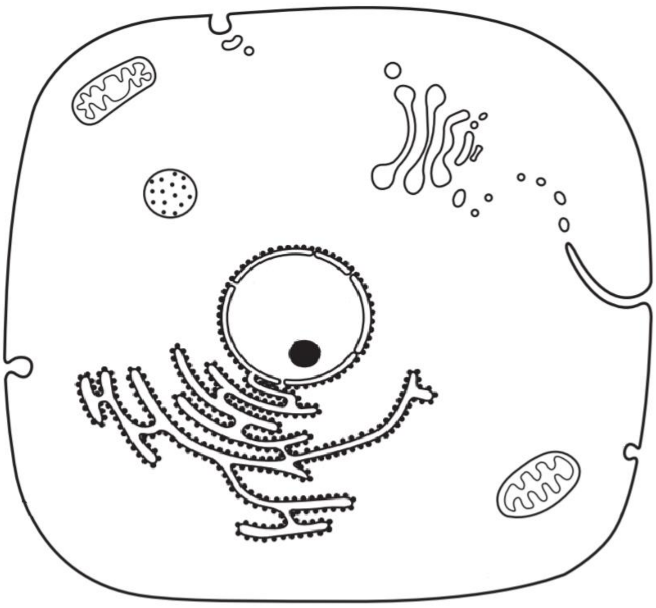

Fig. 3.1

(d) Explain the functions of the Twinkle protein during the replication of mtDNA.

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(e) The *TWNK* gene is also used to make the Twinky protein. While the function of this protein is unknown, some research suggest that it functions similarly to the Twinkle protein.

Explain how the single *TWNK* gene can be used to make two proteins.

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(f) There is broad evidence to indicate that the modern-day mitochondrion is the result of an ancient free-living bacterium which was integrated into an ancestral eukaryotic cell. This is known as the endosymbiont theory.

(i) One evidence for the endosymbiont theory is that the type of DNA possessed by both the mitochondrion and a bacterium is similar.

Apart from this similarity, identify another structural feature that is common to both the mitochondrion and a bacterium.

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(ii) Another evidence for the endosymbiont theory is that the mitochondrion has a double membrane.

Suggest an explanation for how this feature of the mitochondrion supports the endosymbiont theory.

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[Total: 14]

Question 4 starts on page 12

4 Curcumin is a yellow pigment found in the spice turmeric, which is used in curry powder. Curcumin also has medicinal properties.

In vitro experiments show that curcumin decreases the proliferation of cancer cells when cancer cells are cultured in laboratory apparatus.

In one experiment, the mitotic cell cycle of cancer cells in culture was synchronised so that they all entered G_1 phase at the same time. The cells were then transferred to culture media containing different concentrations of curcumin. Nine hours later, the percentage of cells in each phase of interphase was recorded. None of the cells had completed the G_2 phase of interphase.

The results of this experiment are shown in Fig. 4.1.

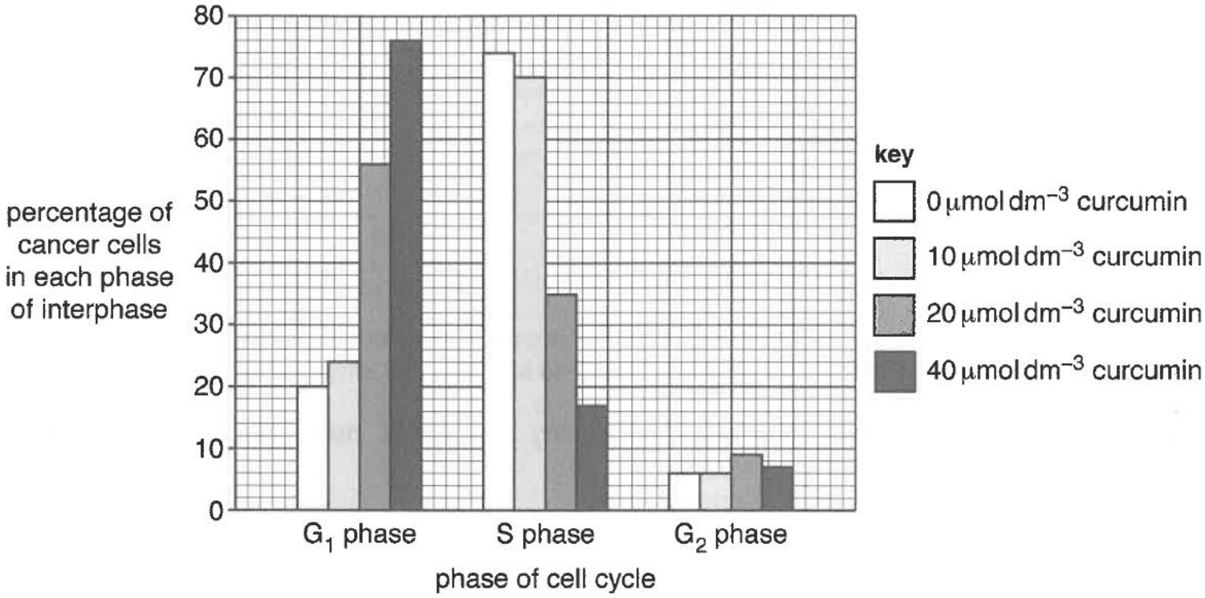

Fig. 4.1

With reference to Fig. 4.1,

(a) describe the effect of curcumin on the percentage of cancer cells in each phase of interphase nine hours after entering G_1 phase.

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(b) suggest how curcumin affects the mitotic cell cycle to prevent cancer cell division.

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[Total: 5]

5 The white campion (*Silene latifolia)* is a flowering plant native to most of Europe, Western Asia and Northern Africa. Male plants are XY and female plants are XX.

In 1912, Baur described the first example of sex-linkage in plants, in *Silene latifolia*.

- An unusual form of *Silene latifolia* with narrow leaves was discovered, and it was found to be a male plant.
- Pollen from this plant was used to fertilise a female plant with normal broad leaves, and all the progeny (male and female) had normal broad leaves.
- When these normal broad-leaved plants were cross-fertilised, 167 plants with broad leaves and 60 with narrow leaves were produced.
- When the narrow-leaved plants were grown to maturity, all were male, while of the broad-leaved plants, 56 were male and 111 were female.

(a) Using the symbols **A** and **a** for leaf shape, draw genetic diagrams to fully illustrate the crosses above.

[5]

(b) All the narrow-leaved plants in the crosses above are male.

Describe a cross in which a female narrow-leaved plant will be produced.

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[Total: 6]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)**, as indicated in the question.

6 **(a)** Compare the structures and functions of tRNA and rRNA. [6]

(b) Discuss the role of proteins in the movement of substances, in biological cells **and** in multicellular organisms. [9]

[Total:15]

7 **(a)** Compare the properties and functions of embryonic stem cells and blood stem cells. [6]

(b) Discuss the significance of hydrophobic interactions in biological molecules **and** structures. [9]

[Total:15]